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Roll No

CM-801(A)-GS

B.E. VIII Semester

Examination, December 2020

Grading System (GS)

Process Piping Design

(Elective - III)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain various requirements of codes and standard in piping system.
b) What is the difference between pipes and tubes?
2. Explain calculation of pressure losses for two-phase flow based on the method of Lockhart-Martinelli.
3. Discuss the following vertical Flow patterns:
Bubble flow, Slug flow, Transition flow, Mist flow.
4. Explain how Rabinowitsch-Mooney equation shows that the wall shear rate for a non-Newtonian fluid can be calculated from the value for a Newtonian fluid having the same flow rate in the same pipe.
5. a) Classify multiphase flow according to flow patterns or regimes.
b) Explain the prediction of frictional pressure gradient for two phase flow models using a concept of Lockhart-Martihelli Model.

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6. Explain the Buckingham's equation for the Bingham-plastic fluids.
7.
 - a) What is film inversion? When is it encountered in two phase flow?
 - b) With the help of neat sketches describe a mixed flow pattern (neither separated nor dispersed) for:
 - i) gas-liquid flow.
 - ii) liquid-liquid flow.
8. Air under standard conditions flows through a 4.0-mm-diameter drawn tubing with an average velocity of $V = 50$ m/s. For such conditions the flow would normally be turbulent. However, if precautions are taken to eliminate disturbances to the flow (the entrance to the tube is very smooth, the air is dust free, the tube does not vibrate, etc.), it may be possible to maintain laminar flow.
 - a) Determine the pressure drop in a 0.1-m section of the tube if the flow is laminar.
 - b) Repeat the calculations if the flow is turbulent.
